The Reflection-Symmetric p-Representation of the Riemann Zeta Logarithmic Derivative

A Unified Framework linking
the Ghosh Function, the Smaller Ghosh Function,
the Completed Xi-Function, and Digamma Corrections

Lord Soumadeep Ghosh et al. with ChatGPT (OpenAI)

Kolkata, India

Abstract

This treatise records an exact reflection-symmetric representation of a logarithmic-derivative combination of the Riemann zeta function. The construction combines the functional equation of the completed xi-function with the substitution $p = s(1-s)$, yielding two reflected branches over $0 < p \leq 1/4$. The result is an algebraic reparameterization of classical identities, not a new criterion for the Riemann Hypothesis. Connections to other disciplines are identified as analogies or possible research directions, not consequences of the identity.

The treatise ends with "The End"

1 Scope and status

The purpose is to state the identity, establish its assumptions, and provide a reproducible computational route. Standard background on the zeta function and its functional equation can be found in [1, 2, 3]. The **Ghosh function** $G(s)$ and the **Smaller Ghosh function** $g(s)$ below are the names assigned here to the defined expressions respectively and should become a standard terminology in analytic number theory.

Throughout, s is real with $0 < s < 1$. On this interval, $\zeta(s) < 0$, so $\zeta(s)\zeta(1-s) > 0$, and the positive square root below is well-defined.

2 Definitions and reflection symmetry

Define

$$P(s) = \sqrt{\zeta(s)\zeta(1-s)} \tag{1}$$

and

$$G(s) = \frac{P'(s)}{P(s)} = \frac{1}{2}\left[\frac{\zeta'(s)}{\zeta(s)} - \frac{\zeta'(1-s)}{\zeta(1-s)}\right]. \tag{2}$$

Consequently,

$$G(1-s) = -G(s), \qquad G\left(\frac{1}{2}\right) = 0. \tag{3}$$

The completed Riemann xi-function is

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \tag{4}$$

and it satisfies

$$\xi(1-s) = \xi(s). \tag{5}$$

Set

$$X(s) = \frac{\xi'(s)}{\xi(s)}, \qquad g(s) = \frac{X(s) - X(1-s)}{2}. \tag{6}$$

Logarithmic differentiation of the functional equation gives $X(1-s) = -X(s)$, hence

$$g(s) = X(s) = \frac{\xi'(s)}{\xi(s)}, \qquad g(1-s) = -g(s), \qquad g\left(\frac{1}{2}\right) = 0. \tag{7}$$

3 The combined identity

Logarithmically differentiating (4) gives

$$\frac{\zeta'(s)}{\zeta(s)} = g(s) - \frac{1}{s} - \frac{1}{s-1} + \frac{1}{2}\log\pi - \frac{1}{2}\psi\left(\frac{s}{2}\right), \tag{8}$$

where $\psi(z) = \Gamma'(z)/\Gamma(z)$ is the digamma function. Replacing s by $1-s$, subtracting, and using the antisymmetry of g, yields

$$\boxed{\begin{aligned} G(s) = g(s) &+ \frac{1}{1-s} - \frac{1}{s} \\ &+ \frac{1}{4}\left[\psi\left(\frac{1-s}{2}\right) - \psi\left(\frac{s}{2}\right)\right]. \end{aligned}} \tag{9}$$

Equivalently,

$$G(s) = g(s) + \frac{2s-1}{s(1-s)} + \frac{1}{4}\left[\psi\left(\frac{1-s}{2}\right) - \psi\left(\frac{s}{2}\right)\right]. \tag{10}$$

4 The p-parameter and two branches

Introduce

$$p = s(1-s). \tag{11}$$

For real $0 < s < 1$, its range is $0 < p \le 1/4$. Solving $s^2 - s + p = 0$ gives

$$s_\pm(p) = \frac{1 \pm \sqrt{1-4p}}{2}. \tag{12}$$

Write $d(p) = \sqrt{1-4p}$. Then

$$s_\pm(p) = \frac{1 \pm d}{2}, \qquad 1 - s_\pm(p) = \frac{1 \mp d}{2}, \tag{13}$$

$$2s_\pm(p) - 1 = \pm d, \qquad \frac{2s_\pm(p) - 1}{p} = \pm\frac{d}{p}. \tag{14}$$

Substitution into (10) gives the combined two-branch representation:

$$\boxed{\begin{aligned} G(s_\pm(p)) &= g(s_\pm(p)) \pm \frac{\sqrt{1-4p}}{p} \\ &\quad \mp \frac{1}{4}\left[\psi\left(\frac{1+\sqrt{1-4p}}{4}\right) - \psi\left(\frac{1-\sqrt{1-4p}}{4}\right)\right], \\ s_\pm(p) &= \frac{1 \pm \sqrt{1-4p}}{2}, \qquad 0 < p \le \frac{1}{4}. \end{aligned}} \tag{15}$$

The upper signs correspond to $s_+(p)$; the lower signs correspond to $s_-(p)$. Since $s_-(p) = 1 - s_+(p)$, reflection symmetry implies

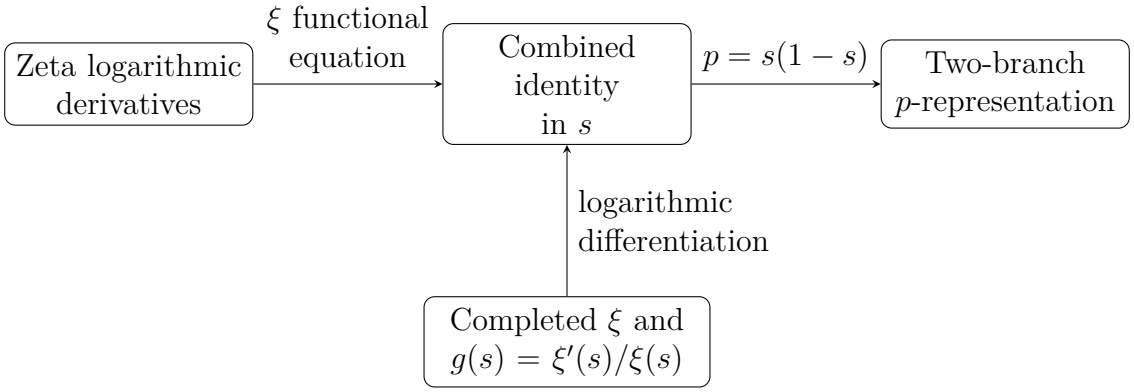
$$G(s_-(p)) = -G(s_+(p)), \qquad g(s_-(p)) = -g(s_+(p)). \tag{16}$$

At $p = 1/4$, both branches coincide at $s = 1/2$, and the rational and digamma correction terms vanish. Thus

$$G\left(\frac{1}{2}\right) = g\left(\frac{1}{2}\right) = 0. \tag{17}$$

5 Structural diagram

The arrows below denote algebraic transformations, not independent theorems.



6 Interdisciplinary interpretation

The symmetry-adapted parameter p labels the pair s and $1 - s$, while G and g are antisymmetric. This resembles symmetry reduction in mathematical modelling. Economics and finance may use invariant state variables; econometrics and data science would require explicit data, estimands, and validation; physics may offer analogies to transformations; computer science can implement the formula as a deterministic algorithm. These are methodological analogies only. The identity itself establishes no results in economics, financial planning, law, chemistry, biology, linguistics, psychology, psychiatry, philosophy, religion, political science, geopolitics, diplomacy, international relations, welfare economics, warfare economics, or cybersecurity. Applications in those fields would require their own assumptions and evidence.

Table 1: Mathematical roles of the principal objects

Object	Definition	Role
$G(s)$	Equation (2)	Antisymmetric zeta combination
$\xi(s)$	Equation (4)	Completed zeta function
$g(s)$	$\xi'(s)/\xi(s)$	Completed-function logarithmic derivative
p	$s(1-s)$	Reflection-invariant parameter
$s_\pm(p)$	Equation (12)	Reflected inverse branches
ψ	Γ'/Γ	Digamma correction

7 Computational verification

A numerical implementation can check the identity at sample points. Such checks detect coding and sign errors but do not replace the derivation.

```
Input: p satisfying 0 < p <= 1/4
d  <- sqrt(1 - 4*p)
s+ <- (1 + d)/2
s- <- (1 - d)/2

For s in {s+, s-}:
    G_direct <- 1/2 * (zeta'(s)/zeta(s)
                       - zeta'(1-s)/zeta(1-s))
    g_value  <- xi'(s)/xi(s)
    correction <- (2*s - 1)/p
                  + 1/4 * (psi((1-s)/2) - psi(s/2))
    G_repr <- g_value + correction
    compare G_direct and G_repr at chosen precision

Check:
    G(s-) = -G(s+)
    g(s-) = -g(s+)
```

Near $p = 0$, the rational term becomes large; near $p = 1/4$, direct subtraction in the digamma difference may lose numerical precision. Higher precision or a stable local expansion can help.

8 Limitations and conclusion

The p-representation is an exact reparameterization of classical identities. It does not by itself provide a new zero-free region, a criterion for the Riemann Hypothesis, or a result about the de Bruijn–Newman constant. Further claims would require independent proofs.

The derivation has three ingredients: the definition of G, the functional equation of the completed xi-function, and the reflection-invariant substitution $p = s(1-s)$. The result organizes the completed-xi contribution and the rational and digamma corrections into one two-branch formula.

References

[1] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.

[2] E. C. Titchmarsh, revised by D. R. Heath-Brown, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986.

[3] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications, vol. 53, American Mathematical Society, 2004.

[4] F. W. J. Olver et al., eds., *NIST Handbook of Mathematical Functions*, Cambridge University Press and National Institute of Standards and Technology, 2010.

Glossary

Completed xi-function
: The function $\xi(s)$ in (4), satisfying $\xi(1-s) = \xi(s)$.

Digamma function
: The logarithmic derivative $\psi(z) = \Gamma'(z)/\Gamma(z)$.

Reflection symmetry
: The transformation $s \mapsto 1 - s$.

Antisymmetric function
: A function f satisfying $f(1-s) = -f(s)$.

Reflection-invariant parameter
: The quantity $p = s(1-s)$, unchanged by reflection.

Two-branch representation
: The pair $s_\pm(p)$ associated with a single value of p.

Logarithmic derivative
: The ratio $f'(s)/f(s)$, defined where $f(s) \neq 0$.

The End